TEST PLAN — SUP-1

Title: Verify that with no headers, in the soap request, error should come to user with 200 OK message

■■ STATUS: DRAFT — PENDING HUMAN REVIEW
This test plan was generated via Advanced Test Plan Creator Skill. It is NOT approved until a QA owner/tester signs off on the Human Review Gate (Section 8).

Jira Issue Key	**SUP-1**	**Issue Type**	[System] Incident
Project	Support (SUP)	**Priority / Severity**	Medium
Status	Open (To Do)	**Reporter / Author**	Vijay Gururaja
Target Component	SOAP Web Services / Gateway	**Attachments**	image-20261029-095606.png (Captured)

1. Scope and Objective

Context: Ticket SUP-1 reports an incident/requirement regarding SOAP request processing. When an incoming SOAP request contains no headers, the service must return a graceful error response with HTTP 200 OK status code.
In Scope:
• Verification of SOAP request payloads arriving without HTTP headers (e.g., missing SOAPAction, Content-Type).
• Verification of SOAP XML Envelopes with missing or empty SOAP Header elements (<soap:Header>).
• Verification that the HTTP response code is strictly 200 OK as specified.
• Validation of error payload structure (SOAP Fault vs. custom error XML/JSON) and user-facing clarity.
• Security inspection to ensure missing-header fault responses do not leak internal stack traces or server paths.
• Baseline verification that valid SOAP requests with complete headers continue to succeed (Regression protection).
Out of Scope:
• Unrelated REST API endpoints or UI-based workflows.
• Long-duration stress or distributed load testing beyond single-node concurrency checks.
Testing Objective: Prove that the SOAP service handles missing headers predictably and safely, returning HTTP 200 OK with an informative error payload, without crashing the application container or exposing sensitive architectural details.

2. Requirement Gap-Analysis Checklist (references/requirement-checklist.md)

Evaluation score: ■ **Present** · ■■ **Ambiguous** · ■ **Missing**. All ambiguous and missing items are converted into explicit questions for the author.

Category	Checklist Item	Score	Tester Observation & Findings
Functional	Clear user story / goal	■ **Missing**	No 'As a / I want / So that' narrative. Only a single summary line exists.
Functional	Testable acceptance criteria	■ **Missing**	Zero formal Acceptance Criteria provided in the Jira ticket.
Functional	Happy path fully described	■■ **Ambiguous**	Normal valid request with complete headers is unstated.
Functional	Negative / error paths described	■■ **Ambiguous**	States error should come with 200 OK, but error schema/code is not defined.
Functional	Boundary & empty states	■ **Missing**	Missing distinction between HTTP headers vs SOAP XML Envelope headers.
Data/Env	Required test data / payload specified	■ **Missing**	No sample SOAP request XML, WSDL, or endpoint URL provided.
Data/Env	Target environment named	■ **Missing**	Not specified which environment (Dev, QA, Staging) hosts the service.
Non-Functional	Security / Information disclosure	■■ **Ambiguous**	Risk of unhandled NullPointerException exposing Java/server stack traces.
Cross-Cutting	Regression / Gateway impact	■■ **Ambiguous**	Returning 200 OK for errors violates standard SOAP 1.1/1.2; client impact unstated.

Clarity	Unambiguous terminology	■■ **Ambiguous**	'no headers' & 'user' ambiguous; typo 'messagea' in attachment.

3. Gaps and Questions for the Ticket Owner (Author: Vijay Gururaja)

"A tester's greatest leverage is asking the decisive question before the bug ships."
The following questions must be answered by the author or product team to unblock execution:

#	Area	Finding (■■/■)	Question to Author (Vijay Gururaja)
Q1	**Header Definition**	■ Ambiguous Scope	Does **'no headers'** mean: a) Missing HTTP request headers (e.g., missing SOAPAction or Content-Type: text/xml)? b) Missing SOAP Envelope Header element (omitted <soap:Header> or empty <soap:Header/>)? c) Missing WS-Security authentication headers (e.g., <wsse:Security>)?
Q2	**Error Payload Schema**	■ Missing Spec	What is the exact response schema expected when headers are omitted? • Is it a standard <soap:Fault> with faultcode and faultstring? • Or a custom application XML tag (e.g. <ErrorResponse>)? • What exact error code and human-readable error message string must be returned?
Q3	**HTTP 200 vs 4xx/500**	■■ Standard Violation	SOAP 1.1 / 1.2 standards specify HTTP 500 for SOAP Faults, and HTTP 400 for malformed client requests. Is returning **HTTP 200 OK** mandatory for legacy ESB/client compatibility? Will downstream consumers parse HTTP 200 responses as success if they rely on HTTP status codes?
Q4	**Target Endpoint & Operation**	■ Missing Spec	What is the exact Service Name, WSDL URL, endpoint path, and operation/action under test? Which testing environment (QA, Staging) has this fix deployed?
Q5	**Security & Logging**	■■ Security Risk	Must we verify that no Java runtime exceptions or server stack traces are leaked inside the 200 OK response payload? Should missing headers trigger a warning log with a correlation/tracking ID in the server logs?

4. Test Scenarios (P0 / P1 / P2 Traceability Matrix)

Derived from ticket summary, edge-case analysis, and security best practices. Each scenario maps directly to the acceptance criteria or a surfaced gap.

ID	Pri	Type	Test Scenario & Steps	Expected Result	Traceability
TS-01	**P0**	Negative	**Omit SOAP Envelope Header element** Send SOAP request containing only <soap:Body> with no <soap:Header> tag.	HTTP 200 OK returned; response body contains well-formed error XML clearly explaining missing header; server does not crash.	Ticket SUP-1 Core requirement
TS-02	**P0**	Negative	**Empty SOAP Header tag** Send SOAP request with self-closing <soap:Header/> or empty tags <soap:Header></soap:Header>.	HTTP 200 OK returned; response contains proper error message indicating required header elements are missing.	Gap Q1 / Boundary
TS-03	**P0**	Negative	**Missing HTTP SOAPAction Header** Send HTTP POST request omitting the HTTP header 'SOAPAction' entirely.	Verify service behavior: returns HTTP 200 OK with error payload (or validates HTTP-level handling without unhandled 500 error).	Gap Q1 / HTTP Headers
TS-04	**P0**	Negative	**Missing or Invalid Content-Type** Send SOAP POST request omitting 'Content-Type' header or sending text/plain.	Verify graceful error handling with 200 OK (or configured gateway protocol error) without raw container fault.	Gap Q1 / HTTP Headers
TS-05	**P0**	Positive	**Regression Baseline: Valid Request with All Headers** Send standard SOAP request with complete HTTP and XML headers.	HTTP 200 OK returned with valid business payload; confirms fix for missing headers did not break valid traffic.	Regression Protection
TS-06	**P1**	Boundary	**Partial / Malformed Header elements** Send SOAP Header with missing required sub-elements (e.g. auth token or transaction ID).	HTTP 200 OK returned; specific validation message stating which sub-element was missing.	Gap Q1 / Q2
TS-07	**P1**	Security	**Security & Information Leakage Audit** Inspect response body when headers are omitted.	Zero stack traces, zero internal IP addresses, zero framework/library versions exposed in response body.	Gap Q5 / Security Guardrail
TS-08	**P1**	Validation	**XML Well-formedness & Schema Compliance** Validate response XML against SOAP WSDL/XSD schema.	Response XML is strictly well-formed, parsable by standard XML parsers without parsing syntax errors.	Gap Q2 / Interoperability

TS-09	**P2**	Load	**Concurrent Missing-Header Requests** Send 50 concurrent requests with missing headers to test thread safety.	All 50 requests return HTTP 200 OK error response consistently; no deadlocks or socket leaks.	Non-functional / Stability
TS-10	**P2**	Observability	**Server Logging & Audit Verification** Check server/gateway logs upon receiving headerless request.	Log entry generated at WARN/INFO level with unique Request ID; no sensitive credentials logged.	Gap Q5 / Observability

5. Test Data and Environment

Test Artifacts & Payloads:
• *Payload A (Valid Baseline):* Full SOAP Envelope containing valid <soap:Header> and authentic credentials.
• *Payload B (No Header Element):* SOAP Envelope containing only <soap:Body> with valid operation payload.
• *Payload C (Empty Header Element):* SOAP Envelope containing <soap:Header/>.
• *Payload D (Malformed/Partial Header):* SOAP Header with missing required sub-tags.
Test Execution Tools: Postman, SoapUI, curl, and Python automated test scripts.
Environment Configuration: Dedicated QA / Staging SOAP service instance with application server logging enabled.

6. Risks and Assumptions

Assumptions Made:
1. Assumed the service protocol is SOAP 1.1 or SOAP 1.2 over HTTP POST.
2. Assumed the requirement for HTTP 200 OK with an error payload is an intentional architectural decision (e.g. legacy enterprise ESB requirement).
3. Assumed a representative test endpoint and sample WSDL will be provided by the development team.
Risks Identified:
• *Client Integration Failure:* Modern API gateways and client SDKs often treat HTTP 200 as successful execution and may fail while attempting to deserialize the business response.
• *Security Masking:* Masking errors under HTTP 200 OK might evade certain Web Application Firewall (WAF) or API monitoring anomaly rules unless content inspection is enabled.
• *Unhandled NullPointer:* Code inspecting headers without null-checks could crash backend worker threads.

7. Entry and Exit Criteria

Entry Criteria:
1. The ticket author (Vijay Gururaja) clarifies and signs off on the open questions in Section 3.
2. Target QA/Staging SOAP service deployed with the latest code build.
3. WSDL contract and sample request schemas provided to QA.
Exit Criteria:
1. 100% of P0 test scenarios (TS-01 through TS-05) executed and passing.
2. Zero security vulnerabilities or raw stack traces exposed in responses (TS-07 passed).
3. No regression observed on standard valid requests with complete headers.
4. Human QA Sign-off recorded on the final Test Plan and execution summary.

■ 8. HUMAN REVIEW GATE (MANDATORY APPROVAL)

Important: In accordance with the *Advanced Test Plan Creator Skill* workflow rules, this document is a **DRAFT**. The AI assistant must STOP here and NOT proceed to test case implementation or test automation until the human QA owner approves or edits the plan.

Summary of What Was Assumed:
• Assumed 'no headers' includes both missing SOAP XML headers and missing HTTP request headers.
• Assumed the requirement for HTTP 200 OK is mandatory despite being non-standard for SOAP faults.
• Assumed target service is SOAP 1.1/1.2 running in QA environment.

Summary of What Could NOT Be Confirmed:
• The exact error response XML schema/format (SOAP Fault vs Custom XML).
• The exact service endpoint, WSDL location, and operation name.
• The reason for requiring HTTP 200 OK instead of standard HTTP 500/400.

Open Questions Blocking Final Sign-off:
• Refer to Section 3: Questions Q1, Q2, Q3, and Q4 require answers from Vijay Gururaja.

■ **ACTION REQUIRED:** Please review the drafted test scenarios, assumptions, and open questions. **Approve, or edit, before I continue to writing test cases or test automation.**